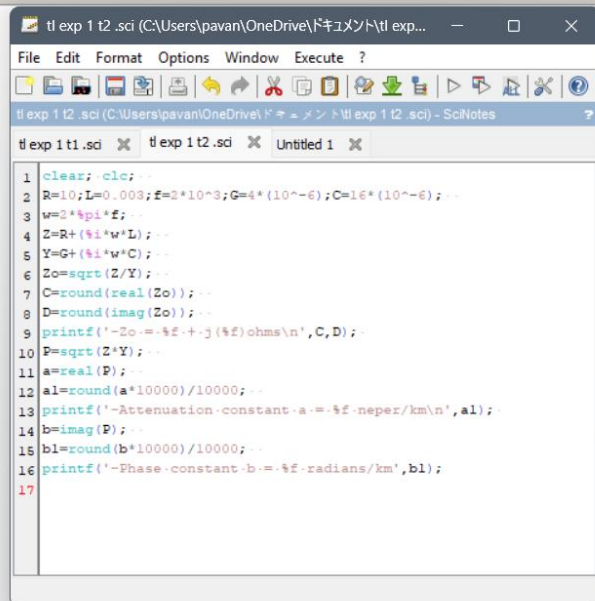


Exp 1 : Estimation of Characteristic Impedance (Z_0), and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test case 1 :

```
clear;
clc;
R=10;
L=0.003;
f=2*10^3;G=4*(10^-6);
C=16*(10^-6);
w=2*pi*f;Z=R+(%i*w*L);
Y=G+(%i*w*C);Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);P=sqrt(Z*Y);
a=real(P);a1=round(a*10000)/10000;printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);b1=round(b*10000)/10000;
printf('-Phase constant b = %f
radians/km',b1);
```

```
Scilab 2025.1.0 Console
Zo = 14.000000 + j(-2.000000) ohms
Attenuation constant a = 0.362100 neper/km
Phase constant b = 2.776900 radians/km
->
```



```
tl exp 1 t2 .sci (C:\Users\pavan\OneDrive\ドキュメント\tl exp...
File Edit Format Options Window Execute ?
tl exp 1 t2 .sci (C:\Users\pavan\OneDrive\ドキュメント\tl exp 1 t2 .sci) - SciNotes
tl exp 1 t1 .sci x tl exp 1 t2 .sci x Untitled 1 x
1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(%i*w*L);
5 Y=G+(%i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('-Zo = %f + j(%f)ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 a1=round(a*10000)/10000;
13 printf('-Attenuation constant a = %f neper/km\n',a1);
14 b=imag(P);
15 b1=round(b*10000)/10000;
16 printf('-Phase constant b = %f radians/km',b1);
17
```

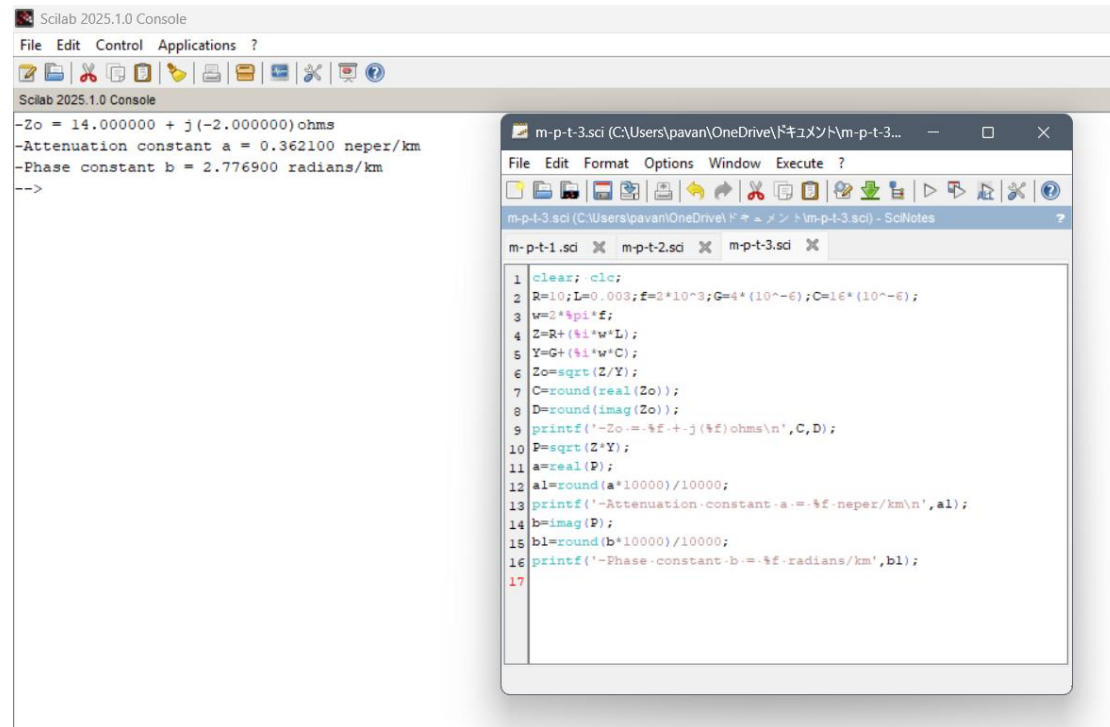
Test case 2 :

```
clear;
clc;
R=10;L=0.003;
f=2*10^3;G=4*(10^-6);
C=16*(10^-6);
```

```

w=2*%pi*f; Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y); C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y); a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1); b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);

```



Test case 3:

```
clear;
```

```
clc;
```

```
R=10;L=0.003;
```

```
f=2*10^3;G=4*(10^-6);C=16*(10^-6);
```

```
w=2*%pi*f;
```

```
Z=R+(%i*w*L);
```

```
Y=G+(%i*w*C);
```

```

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

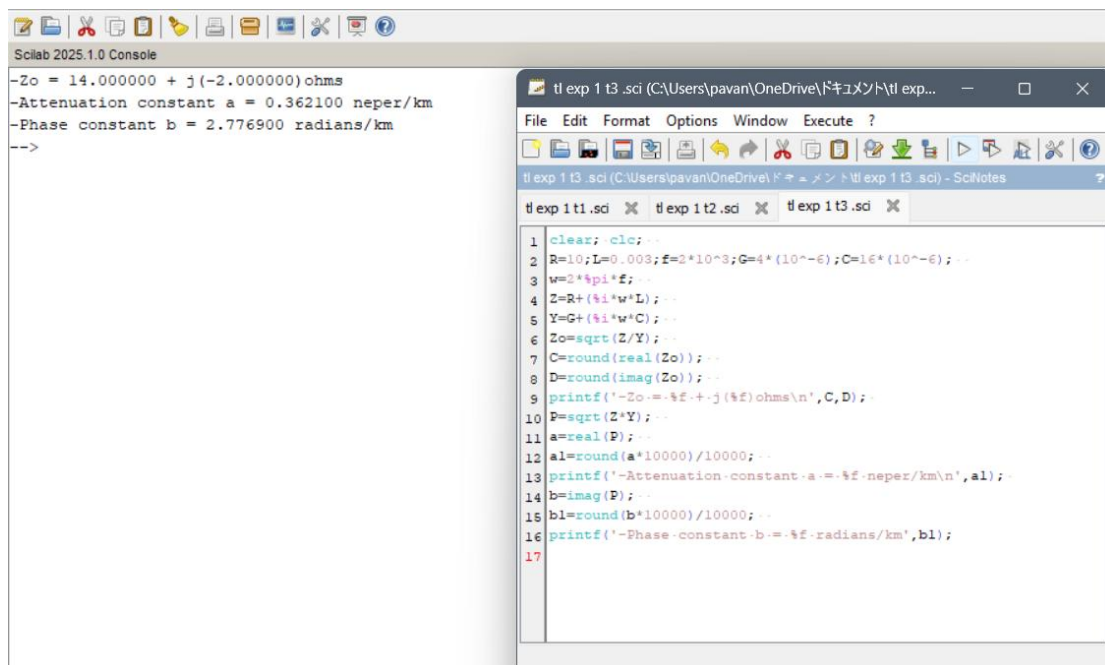
printf('-Attenuation constant a = %f neper/km\n',a1);

b=imag(P);

b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);

```



The screenshot shows the Scilab 2025.1.0 interface. The console window on the left displays the output of the script, and the editor window on the right shows the source code.

Scilab 2025.1.0 Console

```

-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->

```

tl exp 1 t3 .sci (C:\Users\pavan\OneDrive\ドキュメント\tl exp 1 t3 .sci) - SciNotes

```

1 clear;clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(%i*w*L);
5 Y=G+(%i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('-Zo=-%f+-%j(%f)ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 a1=round(a*10000)/10000;
13 printf('-Attenuation constant a=-%f neper/km\n',a1);
14 b=imag(P);
15 b1=round(b*10000)/10000;
16 printf('-Phase constant b=-%f radians/km',b1);
17

```